



PROGRAM

The times indicated are in "Eastern Daylight Time - EDT (Washington DC)"

PROGRAM OVERVIEW (./PROGRAM) SAT, 24 JULY (./SATURDAY) | **SUN, 25 JULY** | MON, 26 JULY (./MONDAY)
| TUE, 27 JULY (./TUESDAY) | WED, 28 JULY (./WEDNESDAY) | THU, 29 JULY (./THURSDAY)

08:00 – 10:00 (sunday#01)

10:30 – 12:30 (sunday#23)

08:00 – 10:00 (EDT - Washington DC)

Human-Computer Interaction

S029: Human-centered Persuasive Computing

Chair(s): Guido Kempter, *Austria*



- **Designing for App Usage Motivation to Support a Gluten-Free Diet by Comparing Various Persuasive Feedback Elements**
Katrin Paldan, Andreas Künz, Walter Ritter, *Austria*; Daire O. Broin, *Ireland*
- **Holdable Devices: Supporting Mindfulness, Psychological Autonomy and Self-Regulation During Smartphone Use**
Federico Julien Tiersen, Rafael Alejandro Calvo, *United Kingdom*
- **Annoyed to Discontinue: Factors Influencing (Dis)Continuance of Using Activity Tracking Wearables**
Kaja J. Fietkiewicz, Aylin Ilhan, *Germany*
- **Better Performance Through Mindfulness: Mobile Application Design for Mindfulness Training to Improve Performance in College Athletes**
Felicia Roger-Hogan, Tayler Wullenweber, Jung Joo Sohn, *United States*
- **Designing for Self-Awareness: Evidence-based Explorations of Multimodal Stress-Tracking Wearables**
Riccardo Chianella, Marco Mandolfo, Riccardo Lolatto, Margherita Pillan, *Italy*

S030: Facial Expressions and Emotions

Chair(s): Jinhui Chao, *Japan*



- **A New Algorithm to Find Isometric Maps for Comparison and Exchange of Facial Expression Perceptions**
Masashi Shinto, Jinhui Chao, *Japan*
- **The Impact of Facial Attractiveness and Affective Person Knowledge on Visual Awareness**
Junchen Shang, Hong Yang, *P.R. China*
- **PerformEyebrow: Design and Implementation of an Artificial Eyebrow Device Enabling Augmented Facial Expression**
Motoyasu Masui, Yoshinari Takegawa, Nonoka Nitta, Yutaka Tokuda, Yuta Sugiura, Katsutoshi Masai, Keiji Hirata, *Japan*
- **An Experimental Study on “Consensus to Match” Game for Analyzing Emotional Interaction in Consensus Building Process**
Kyoko Ito, Yoshiki Sakamoto, Rieko Yamamoto, Mizuki Yamawaki, Daisuke Miyazaki, Kimi Ueda, Hirotake Ishii, Hiroshi Shimoda, *Japan*

S031: Interacting with Small and Tiny Screens

Chair(s): Sayan Sarcar, *Japan*

- **One Stroke Alphanumeric Input Method by Sliding-in and Sliding-out on the Smartwatch Screen**
Toshimitsu Tanaka, Hideaki Shimazu, Yuji Sagawa, *Japan*
- **Investigation of Motion Video Enhancement for Image-based Avatars on Small Displays**
Tsubasa Miyauchi, Wataru Ganaha, Masashi Nishiyama, Yoshio Iwai, *Japan*
- **Character Input Method Working on 1-inch Round Screen for Tiny Smartwatches**
Ojiro Suzuki, Toshimitsu Tanaka, Yuji Sagawa, *Japan*
- **A Study on the Usability of Different Age Groups to the Interface of Smart Bands**
Xiao-Yu Jia, Chien-Hsiung Chen, *Taiwan*, China*
- **Current Status of User Experience of the Keyboard on Smartphones: An Overall Questionnaire Analysis**
Yincheng Wang, Tong Lin, Jingxin Yu, Lu Wang, Jibo He, Luoma Ke, *P.R. China*



- **The Crowd Thinks Aloud: Crowdsourcing Usability Testing with the Thinking Aloud Method**
Edwin Gamboa, Rahul Galda, Cindy Mayas, Matthias Hirth, *Germany*
 - **Collaborative Heuristic Inspection: Evaluator Experiences and Organization of Work**
Yavuz Inal, Jo Wake, Frode Guribye, *Norway*
 - **The Hidden Cost of Using Amazon Mechanical Turk for Research**
Antonios Saravanos, Stavros Zervoudakis, Dongnanzi Zheng, Neil Stott, Bohdan Hawryluk, Donatella Delfino, *United States*
 - **Is Usability Evaluation of DSL Still a Trending Topic?**
Ildevana Poltronieri, Allan Christopher Pedroso, Avelino Francisco Zorzo, Maicon Bernardino, Marcia De Borba Campos, *Brazil*
 - **Challenges and workarounds of conducting Augmented Reality Usability tests remotely a case study**
Ted Kim, Santiago Arconada Alvarez, Young-Mi Choi, *United States*
 - **User Assessment of Webpage Usefulness**
Ning Sa, Xiaojun Yuan, *United States*
-

Human Interface and the Management of Information

S033: Data, Semantics and Interaction

Chair(s): Yumi Asahi, *Japan*



- **Classification of Automotive Industry Salesmen**
Yoshio Matsuyama, Yumi Asahi, *Japan*
- **Evaluation of the Current State of Nippon Professional Baseball in Digitalization**
Masaru Kondo, Yumi Asahi, *Japan*
- **Proposal of Credit Risk Model using Machine Learning in Motorcycle Sales**
Ryota Fujinuma, Yumi Asahi, *Japan*

Universal Access in Human-Computer Interaction

S034: Emotion and User Experience in HCI

Chair(s): Silvia Ceccacci, Maura Mengoni, *Italy*

- **Affective Guide for Museum: a System to Suggest Museum Paths based on Visitors' Emotions**
Alex Altieri, Silvia Ceccacci, Luca Giraldi, Alma Leopardi, Maura Mengoni, *Italy*; Abudukaiyoumu Talipu, *P.R. China*
- **Building an Ecologically Valid Facial Expression Database – Behind the Scenes**
Francesca Nonis, Luca Ulrich, Nicolò Dozio, Francesca Giada Antonaci, Enrico Vezzetti, Francesco Ferrise, Federica Marcolin, *Italy*
- **Supervised Contrastive Learning for Game-Play Frustration Detection from Speech**
Meishu Song, *Germany*; Emilia Parada-Cabaleiro, *Austria*; Shuo Liu, Manuel Milling, Alice Baird, Zijiang Yang, Björn Schuller, *Germany*
- **Usability Testing on Tractor's HMI: a Study Protocol**
Elisa Prati, Fabio Grandi, Margherita Peruzzini, *Italy*
- **Geo-education as a Valuable Support to Children with Learning Difficulties**
Marianna Di Gregorio, Monica Sebillio, Giuliana Vitiello, *Italy*
- **Implicit Measures as a Useful Tool for Evaluating User Experience**
Rossana Actis-Grosso, Roberta Capellini, Francesco Ghedin, Francesca Tassistro, *Italy*





Chair(s): Denis Gracanin, *United States*

- **Extended Reality, Pedagogy, and Career Readiness: A Review of Literature**

Patrick Guilbaud, T. Christa Guilbaud, Dane Jennings, *United States*

- **A Review of Distributed VR Co-Design Systems**

Jean-François Lapointe, Norman G. Vinson, Keiko Katsuragawa, Bruno Emond, *Canada*

- **The Mobile Office: A Mobile AR Systems for Productivity Applications in Industrial Environments**

Daniel Antonio Linares Garcia, Poorvesh Dongre, Nazila Roofigari-Esfahan, Doug Bowman, *United States*

- **Usability and User Experience of Interactions on VR-PC, HoloLens 2, VR Cardboard and AR Smartphone in a Biomedical Application**

Manisha Suresh Balani, Johannes Tümler, *Germany*

- **The Potential of Augmented Reality for Remote Support**

Stefan Kohn, Moritz Schaub, *Germany*

- **Supporting Embodied and Remote Collaboration in Shared Virtual Environments**

Mark Manuel, Poorvesh Dongre, Abdulaziz Alhamadani, Denis Gracanin, *United States*

Cross-Cultural Design

S036: CCDHA -- Cross-Cultural Design for Humanity and Arts - I

Chair(s): I-Ying Chiang, Rungtai Lin, *Taiwan**, *China*



- **The Strategies of Experiential Design in the Creative Life Industry**
Shu-Hua Chang, *Taiwan**, *China*
- **A Study of Applying Bauhaus Design Idea into the Reproduction of the Triadic Ballet**
Yiwen Ting, Po-Hsien Lin, Rungtai Lin, *Taiwan**, *China*
- **A Study of Sightseeing Illustration Map Design**
Wai Kit NG, *Hong Kong*; Jing Cao, *Taiwan**, *China*
- **Application of Experience Design in Environmental Education Experience Activities**
Cheng Hsiang Yang, Rungtai Lin, Po-Hsien Lin, *Taiwan**, *China*
- **Brand Construction of Chinese Traditional Handicrafts in the We-media Era—A Case Study of “Rushanming”, a Ru Ware Brand**
Shuang Ou, Minghong Shi, Xin Wen, *P.R. China*; Rungtai Lin, *Taiwan**, *China*
- **Exploring the Integration of Emotion and Technology to Create Product Value — A Case Study on QisDesign Lighting**
Jen-Feng Chen, Po-Hsien Lin, Rungtai Lin, *Taiwan**, *China*

S037: Cross-Cultural Product and Service Design in Intelligent Environment

Chair(s): Runting Zhong, *P.R. China*





- **Cross-cultural Research on Consumer Decision Making of HNB Product Modeling based on Eye Tracking**
Lizhong Hu, Lili Sun, Yong Zha, Min Chen, Lei Wu, Huai Cao, *P.R. China*
- **Identifying Design Feature Factors Critical to Acceptance of Smart Voice Assistant**
Na Liu, Ruoxuan Liu, Wentao Li, *P.R. China*
- **Lost in Interpretation? The Role of Culture on Rating the Emotional Nonverbal Behaviors of a Virtual Agent**
Adineh Hosseinpanah, Nicole C. Krämer, *Germany*
- **What Do Young Adults Like About E-planting? A Questionnaire Survey**
Runting Zhong, Hongyi Fang, Jiahui Rao, Yue Sun, Ji Xu, Xin Peng, *P.R. China*

Social Computing and Social Media

S038: Evolutional Marketing with HCI

Chair(s): Kohei Otake, Takashi Namatame, *Japan*





- **Analysis of User Relationships on Cooking Recipe Site Using Network Structure**
Yuzuki Kitajima, Kohei Otake, Takashi Namatame, *Japan*
- **Prediction for Private Brand Items Purchase Behavior of Hair Salons Using Bayesian Survival Analysis**
Yuho Katagiri, Kohei Otake, Takashi Namatame, *Japan*
- **Verification of Probabilistic Latent Semantic Analysis Clustering Solution Stability and Proposal of Optimal Initial Values Setting Method**
Shinnosuke Terasawa, Kohei Otake, Takashi Namatame, *Japan*
- **Evaluation of Cooking Recipes Using Their Texts and Images**
Mei Nonaka, Kohei Otake, Takashi Namatame, *Japan*
- **Customer Visit Prediction Using Purchase Behavior and Tendency**
Retsuya Saito, Kohei Otake, Takashi Namatame, *Japan*

Augmented Cognition

S039: Augmented Cognition through Immersive Experiences

Chair(s): Ayoung Suh, *Korea*



- **The Impact of Auditory Based Immersive Virtual Travel Experience on Mental Health of the Visually Impaired**
Linghong Li, *P.R. China*
- **Reducing Driver's Cognitive Load with the Use of Artificial Intelligence and Augmented Reality**
Kweku Bram-Larbi, Vassilis Charissis, Ramesh Lagoo, *United Kingdom*; Shu Wang, *P.R. China*; Soheeb Khan, *United Kingdom*; Samar Altarteer, *Saudi Arabia*; David Harrison, *United Kingdom*; Dimitris Drikakis, *Cyprus*
- **A Literature Review on a Neuro-Psychological Approach to Immersive Technology Research**
Guan Wang, *P.R. China*; Ayoung Suh, *Korea*
- **The Cognitive Study of Immersive Experience in Science and Art Exhibition**
Ching Wen Chang, *P.R. China*
- **Automatic Engagement Recognition for Distance Learning Systems: A Literature Study of Engagement Datasets and Methods**
Shofiyati Nur Karimah, *Indonesia*; Shinobu Hasegawa, *Japan*

Digital Human Modeling and Applications in Health, Safety, Ergonomics and Risk Management

S040: Human-Robot-AI Interaction

Chair(s): Yaqin Cao, *P.R. China*



- **Influence of HNB Product Packaging Health Warning Design on Risk Perception Based on Eye Tracking**
Lili Sun, Lizhong Hu, Feng Zheng, Yue Sun, Huai Cao, Lei Wu, *P.R. China*
- **An Empirical Study of the Influencing Factors on User Experience for Barrage Video Website – A Case Study of Bilibili**
Weilin Liu, Zhaoshuang He, Mengxin Liu, *P.R. China*
- **The Influence of the Aesthetic Design of Taobao APP on Users' Emotional Experience**
Yimeng Zhang, Yang Zhang, Jiaojiao Gao, *P.R. China*

Design, User Experience, and Usability

S041: Visual Languages: from Sign to Meaning

Chair(s): Cesco Reale, *Switzerland*

- **Visual Writing at the State-of-the-Art?**
Jochen Gros, *Germany*
- **YERKISH: A Visual Language for Computer-Mediated Communication by an Ape**
Marco Cesare Bettoni, *Switzerland*
- **Emojitaliano: A Social and Crowdsourcing Experiment of the Creation of a Visual International Language**
Johanna Monti, Francesca Chiusaroli, *Italy*; Federico Sangati, *Japan*
- **From Hieroglyphs to Emoji, to IKON: the Search of the (Perfect?) Visual Language**
Cesco Reale, Marwan Kilani, *Switzerland*; Araceli Gimenez, *Spain*; Nadu Barbashova, *Germany*; Roman Oechsli, *Switzerland*
- **Pasigraphy: Universal Visible Languages**
Aaron Marcus, *United States*



- **Web Analytics for User Experience: A Systematic Literature Review**
Fryda Palomino Diaz, Freddy Paz, Arturo Moquillaza, *Peru*
- **Challenges and Opportunities on the Application of Heuristic Evaluations: A Systematic Literature Review**
Adrian Lecaros, Freddy Paz, Arturo Moquillaza, *Peru*
- **Validation of a Questionnaire to Evaluate the Usability in the Peruvian Context**
Freddy Paz, Freddy Asrael Paz, Arturo Moquillaza, Joel Aguirre, *Peru*; Fiorella Falconi, *United States*; Jaime Diaz, *Chile*; Hilmar Hinojosa, *Peru*
- **A Systematic Review of User-Centered Design Techniques Applied to the Design of Mobile Application User Interfaces**
Percy Quezada, Rony Cueva, Freddy Paz, *Peru*
- **Relations on Cultural Behavior and Technology Adoption: A Chilean Perspective**
Jaime Diaz, Danay Ahumada, Jorge Hochstetter, *Chile*; Freddy Paz, *Peru*
- **Accessibility Evaluation of E-Government Web Applications: A Systematic Review**
Daniela Cisneros, Fernando Huamán Monzón, Freddy Paz, *Peru*

Distributed, Ambient and Pervasive Interactions

S043: Interaction in Intelligent Environments - II

Chair(s): Heinrich Ruser, *Germany*





- **Re-Imagining Indoor Space Utilization in the COVID-19 Pandemic with Smart Re-configurable Spaces (SReS)**
Poorvesh Dongre, Mark Manuel, Denis Gracanin, *United States*
- **Touchless Interaction on Mobile Devices using Embedded Ambient Light Sensor**
Alen Salkanovic, Sandi Ljubic, *Croatia*
- **Home Appliance Control Using Smartwatches with Continuous Gesture Recognition**
Thamer Horbylon Nascimento, Fabrizzio Soares, *Brazil*
- **“Point at it with your smartphone”: Assessing the applicability of orientation sensing of smartphones to operate IoT devices**
Heinrich Ruser, *Germany*; Ilan Kirsh, *Israel*

HCI in Business, Government and Organizations

S044: Virtual Customer Interaction

Chair(s): Andrea Müller, *Germany*



- **Haptic Interaction for VR: Use-Cases for Learning and UX, Using the Example of the BMBF project SmartHands**
Oliver Gast, Alina Makhkamova, Dirk Werth, Mareike Funk, *Germany*
- **Customer Solution Design – A New Agile Role Needed in the Automotive Industry to Support Digital Transformation?**
Aline Schnurr, Pamela Renz, Andrea Müller, *Germany*
- **Empirical Research as a Challenge in Day-to-Day Teaching during the Pandemic of 2020/21 - Practical Solutions**
Christina Miclau, Annebeth Demaeght, Andrea Müller, *Germany*
- **Methodical Framework and Case Study for Empowering Customer-Centricity in an E-Commerce Agency–the Experience Logic as Key Component of User Experience Practices within Agile IT Project Teams**
Beatrice Weber, Andrea Müller, Christina Miclau, *Germany*

Learning and Collaboration Technologies

S045: Interactive Learning Ecosystems - II

Chair(s): David Fonseca, Francisco J. García-Peñalvo, *Spain*





Marian Alaez, Susana Romero, David Fonseca, Daniel Amo, Enric Peña, Silvia Necchi, *Spain*

- **WhatsApp or Telegram. Which is the Best Instant Messaging Tool for the Interaction in Teamwork?**

Miguel A. Conde, Francisco J. Rodríguez-Sedano, Francisco J. Rodríguez-Lera, Alexis Gutiérrez-Fernández, Ángel Manuel Guerrero-Higueras, *Spain*

- **How to Assess Empathy during Online Classes**

Karim Elia Fraoua, *France*

- **Heightmap examiner. A descriptive tool for education in digital cartography.**

Luis Hernández-Ibáñez, Viviana Barneche-Naya, *Spain*

- **A Social, Virtual and Open Model for Measuring Creativity**

Carlos Guillem, Rafael Molina-Carmona, *Spain*

Human Aspects of IT for the Aged Population

S046: Designing for the Elderly - I

Chair(s): Helene Fournier, *Canada*





United Kingdom

- **System Architecture and User Interface Design for a Human-Machine Interaction System for Dementia Intervention**

Miroslava Migovich, Ritam Ghosh, Nibraas Khan, Judith A. Tate, Lorraine M. Mion, Nilanjan Sarkar, *United States*

- **Designing a Dementia-informed, Accessible, Co-located Gaming Platform for Diverse Older Adults with Dementia, Family and Carers**

Paula Gardner, Stephen Surlin, Adekunle Akinyema, Jessica Rauchberg, Caitlin McArthur, Jenny Hao, Rong Zheng, Alexandra Papaioannou, *Canada*

- **Remote User Testing for an Age-friendly Interface Design for Smart Homes**

Adriana Marques Da Silva, Hande Ayanoglu, Bruno Silva, *Portugal*

- **Tangible Chess for Dementia Patients – Playing with Conductive 3D printed Figures on a Touchscreen**

Christian Eichhorn, David A. Plecher, Oleksandr Golovnya, Dorothee Volkert, *Germany*; Atsushi Hiyama, *Japan*; Gudrun Klinker, *Germany*

- **Review of Remote Usability Methods for Aging in Place Technologies**

Irina Kondratova, Helene Fournier, Keiko Katsuragawa, *Canada*

HCI for Cybersecurity, Privacy and Trust

S047: User Identity and Personal Data Protection

Chair(s): Joseph Aneke, *Italy*





- **Identity Recognition based on the Hierarchical Behavior Characteristics of Network Users**
Biao Wang, Zhengang Zhai, Bingtao Gao, Li Zhang, *P.R. China*
- **Understanding User's Behavior and Protection Strategy upon Losing, or Identifying Unauthorized Access to Online Account**
Huzeyfe Kocabas, *Turkey*; Swapnil Nandy, *India*; Tanjina Tamanna, *Bangladesh*; Mahdi Nasrullah Al-Ameen, *United States*
- **Partial Consent: A Study on User Preference for Informed Consent**
Sven Bock, *Germany*; Ashraf Ferdouse Chowdhury, Nurul Momen, *Sweden*

HCI in Games

S048: Recent Developments in Gaming and HCI - I

Chair(s): Khaldoun Dhou, *United States*; Barbara Caci, *Italy*





- **Gameplay as Network: Understanding the Consequences of Automation on Play and Use**
Nicolas LaLone, *United States*
- **Gaze as a Navigation and Control Mechanism in Third-Person Shooter Video Games**
Patricio Isbej, Francisco J. Gutierrez, *Chile*
- **Detecting Real-Time Correlated Simultaneous Events in Microblogs: The case of Men's Olympic Football**
Samer Muthana Sarsam, *Malaysia*; Hosam Al-Samarraie, *United Kingdom*; Nurhidayah Bahar, Abdul Samad Bin Shibghatullah, *Malaysia*; Atef Eldenfria, *Libya*; Ahmed Al-Sa'Di, *New Zealand*
- **Can Games and Gamification Improve Online Learners' Outcomes and Satisfaction on the Madrasati Platform in Saudi Arabia?**
Wad Ghaban, *United Kingdom*
- **Analysis of the Competitiveness of Asymmetric Games in the Market**
Jiawei Dai, Xinrong Li, *P.R. China*

HCI in Mobility, Transport and Automotive Systems

S049: HCI Issues and Assistive Systems for Users with Special Needs in Mobility

Chair(s): Christophe Kolski, Sophie Lepreux, *France*



- **A Causal Model of Intersection-Related Collisions for Drivers With and Without Visual Field Loss**
Bianca Biebl, Severin Kacianka, Anirudh Unni, Alexander Trende, Jochem W. Rieger, Andreas Lüdtke, Alexander Pretschner, Klaus Bengler, *Germany*
- **Assistive Systems for Mobility in Smart City: Humans and Goods**
Yuhang Li, Chuantao Yin, Zhang Xiong, Bertrand David, René Chalon, Hao Sheng, *France*
- **Discussion of Intelligent Electric Wheelchairs for Caregivers and Care Recipients**
Satoshi Hashizume, Ippei Suzuki, Kazuki Takazawa, Yoichi Ochiai, *Japan*
- **Electroencephalography Shows Effects of Age in Response to Oddball Auditory Signals: Implications for Semi-Autonomous Vehicle Alerting Systems for Older Drivers**
Melanie Turabian, Kathleen Van Benthem, Chris Herdman, *Canada*

Adaptive Instructional Systems

S050: Learner State Diagnostics for Adaptive Instruction

Chair(s): Jessica Schwarz, *Germany*





- **Measuring Flow, Immersion and Arousal/Valence for Application in Adaptive Learning Systems**
Ehm Kannegieser, Daniel Atorf, Joachim Herold, *Germany*
- **Taxonomy of Physiologically Adaptive Systems and Design Framework**
John Muñoz, *Canada*; Luis Quintero, *Sweden*; Chad Stephens, Alan Pope, *United States*
- **Measuring Cognitive Load for Adaptive Instructional Systems by Using a Pressure Sensitive Computer Mouse**
Thomas E.F. Witte, Henrike Haase, Jessica Schwarz, *Germany*
- **Individual Differences in the Relationship between Emotion and Performance in Command-and-Control Environments**
Alina Schmitz-Hübsch, Sophie-Marie Stasch, Sven Fuchs, *Germany*

Design, Operation and Evaluation of Mobile Communications

S051: Mobile Systems Users Adoptions

Chair(s): June Wei, *United States*





- **The Impact of Mobile IT on the Service Innovation Performance of Manufacturing**
Caihong Liu, *P.R. China*; Hannah Ji, June Wei, *United States*
- **An Empirical Study on the Impact of Blockchain Technology on the Cost of Company Equity Financing**
Yan Yue, *P.R. China*; Hannah Ji, *United States*; Yirong Ying, *P.R. China*
- **Can Informal Information Transmission Mechanisms Improve the Information Efficiency of the Capital Market? —Based on the Empirical Research of the Executive Community**
Yan Yue, *P.R. China*; Hannah Ji, *United States*; Yirong Ying, Dingfu Jiang, *P.R. China*
- **Transformation the Business of eCommerce through Blockchain**
Uttam Kumar Roy, Weining Tang, *P.R. China*

Culture and Computing

S052: Digital Technologies and Heritage Conservation - I

Chair(s): Claudia Trillo, Busisiwe Chikomborero Ncube Makore, *United Kingdom*



Kwasi Gyau Baffour Awuah, *United Kingdom*; Dilip A. Patel, Lukman E Mansuri, *India*

- **A SLAM Integrated Approach for Digital Heritage Documentation**

Salvatore Barba, Carla Ferreyra, *Italy*; Victoria Andrea Cotella, *Argentina*; Andrea Di Filippo, Secondo Amalfitano, *Italy*

- **Digital Storytelling: the Integration of Intangible and Tangible Heritage in the city of Surat, India**

Chika Udejaja, *United Kingdom*; Lukman E Mansuri, *India*; Busisiwe Chikomborero Ncube Makore, Kwasi Gyau Baffour Awuah, *United Kingdom*; Dilip A. Patel, *India*; Claudia Trillo, *United Kingdom*; K N Jha, *India*

- **Interactive Tools for the Visualization of Tangible and Intangible Silk Heritage Emerging from an Interdisciplinary Work**

Cristina Portalés, Jorge Sebastián, Javier Sevilla, Ester Alba, Marcos Fernández, Mar Gaitán, Pablo Casanova-Salas, Arabella León, Manolo Pérez, Eliseo Martínez Roig, Jesús Gimeno, *Spain*

S053: Visitor Experiences in Digital Culture

Chair(s): To be announced

- **User Experience in Digital Museums: A Case Study of the Palace Museum in Beijing**

Wenhua Li, Jia Xin Xiao, *P.R. China*

- **Indoor Location-based AR for Large-Scale Museum Exhibition**

Sumin Ahn, Dongsoo Han, *Korea*

- **Before You Visit-: New Opportunities for the Digital Transformation of Museums**

Hyungmin Park, Jeongyun Heo, Jongkeon Kim, *Korea*

- **Research on Factors Influencing Users' Technology Acceptance of Virtual Museums**

Wenxuan Gong, Bing Xiao, *P.R. China*

- **Analysis of the Influencing Factors of User Experience in Online Art Forms during the COVID-19—based on the DEMATEL Method**

Chenqi Zhang, Ting Han, Chunrong Liu, *P.R. China*





Chair(s): Pei-Hsuan Hsieh, *Taiwan**, *China*

- **The Challenge of Digital Education and Equality in Taiwan**

Shin-yi Lee, *Taiwan**, *China*

- **Estimation of Learners' Engagement Using Face and Body Features by Transfer Learning**

XianWen Zheng, *P.R. China*; Shinobu Hasegawa, *Japan*; Minh-Tuan Tran, *Viet Nam*; Koichi Ota, Teruhiko Unoki, *Japan*

- **Tool or Partner: The Designer's Perception of an AI-style Generating Service**

Kyungsun Kim, Jeongyun Heo, Sanghoon Jeong, *Korea*

- **Roles of Artificial Intelligence and Extended Reality Development in the Post-COVID-19 Era**

Chutisant Kerdvibulvech, *Thailand*; ZhaoYang Dong, *Australia*

S055: Approaches for Engaging Humans in the AI Lifecycle

Chair(s): Brian C. Stanton, *United States*

- **A Contextual Bayesian User Experience Model for Scholarly Recommender Systems**

Zohreh Champiri, Brian Fisher, *Canada*; Chun Yong Chong, *Malaysia*

- **Collaborative Human-AI Sensemaking for Intelligence Analysis**

Stephen L. Dorton, Robert Hall, *United States*

- **How Intuitive Is It? Comparing Metrics for Attitudes in Argumentation with a Human Baseline**

Markus Brenneis, Martin Mauve, *Germany*

- **Automatic Generation of Machine Learning Synthetic Data using ROS**

Kyle M. Hart, Ari B. Goodman, Ryan P. O'Shea, *United States*



10:30 – 12:30 (EDT - Washington DC)

Human-Computer Interaction

S056: Socio-cultural and Linguistic Aspects in Human-Computer Interaction

Chair(s): Christina Alexandris, *Greece*

- **Graphic Representations of Spoken Interactions from Journalistic Data: Persuasion and Negotiations**
Christina Alexandris, Vasilios Floros, Dimitrios Mourouzidis, *Greece*
 - **A Study on Universal Design of Musical Performance System**
Sachiko Deguchi, *Japan*
 - **Grounding of Concept, Indexical and Name**
Roland Hausser, *Germany*
 - **A Sentiment Analysis Web Platform for Multiple Social Media Types and Language-Specific Customizations**
Stavros Giannakis, Christina Valavani, Christina Alexandris, *Greece*
 - **“Tell me your story, I’ll tell you what makes it meaningful”: Characterization of Meaningful Social Interactions Between Intercultural Strangers and Design Considerations for Promoting Them**
Maria Laura Ramirez Galleguillos, Aya Eloiriachi, Büşra Serdar, Aykut Coşkun, *Turkey*
 - **Evaluating the effectiveness of machine translation of audio description in the English-Dutch language pair**
Nina Reviere, Gert Vercauteren, Kim Steyaert, *Belgium*
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S057: Interaction in Art, Design and Edutainment

Chair(s): Salah Ahmed, *Norway*; Mohammad Shidujaman, *P.R. China*



- **Educational Robot European Cross-Cultural Design**
Anna-Maria Velentza, Stavros Ioannidis, Nefeli Georgakopoulou, *Greece*; Mohammad Shidujaman, *P.R. China*; Nikolaos Fachantidis, *Greece*
- **Affective Robot Learner: Implementation of Artificial Emotion System Inspired by Educational Psychology**
Binnur Gorer, H. Levent Akin, *Turkey*
- **AI Creativity and The Human-AI Co-Creation Model**
Zhuohao Wu, Danwen Ji, Kaiwen Yu, Xianxu Zeng, Dingming Wu, Mohammad Shidujaman, *P.R. China*
- **A Design Method of Children Playground Based on Bionic Algorithm**
Fei Yue, Wenda Tian, Mohammad Shidujaman, *P.R. China*
- **The Smaller the Better? A Study on Acceptance of 3D Display of Exhibits of Museum's Mobile Media**
Xinhao Guo, Jingjing Qiao, Ran Yan, Ziyun Wang, Junjie Chu, *P.R. China*

S058: Supporting Collaboration

Chair(s): Lawrence J. Henschen, *United States*



- **Application for the Cooperative Control of Mobile Robots with Energy Optimization**
José Varela-Aldás, Christian Ichina, Belén Ruales, Víctor H. Andaluz, *Ecuador*
- **Current Problems, Future Needs: Voices of First Responders about Communication Technology**
Kerriane Morrison, Shanee Dawkins, Yee-Yin Choong, Mary Theofanos, Kristen Greene, Susanne Furman, *United States*
- **What Futuristic Technology Means for First Responders: Voices from the Field**
Shanee Dawkins, Kerriane Morrison, Yee-Yin Choong, Kristen Greene, *United States*
- **An Evaluation of Remote Workers' Preferences for the Design of a Mobile App on Workspace Search**
Cátia Carvalho, Edirlei Soares de Lima, Hande Ayanoglu, *Portugal*
- **Digital Communication to Compensate for Social Distancing? - Results of a Survey on the Local Communication App DorfFunk**
Matthias Berg, Anne Hess, Matthias Koch, *Germany*

S059: Technology Acceptance and Trust

Chair(s): Valeria Orso, *Italy*

- **Teens' Conceptual Understanding of Web Search Engines: The Case of Google Search Engine Result Pages (SERPs)**
Dania Bilal, Yan Zhang, *United States*
- **Two-Way Human-Agent Trust Relationships in Adaptive Cognitive Agent, Adaptive Tasking Scenarios: Literature Metadata Analysis**
Daniel Kennedy, Maartje Hidalgo, *United States*
- **Are Professional Kitchens Ready for Dummies? A Comparative Usability Evaluation between Expert and non-Expert Users**
Valeria Orso, Daniele Verí, Riccardo Minato, Alessandro Sperduti, Luciano Gamberini, *Italy*
- **Too Hot to Enter: Investigating Users' Attitudes toward Thermoscanners in COVID times**
Alice Bettelli, Valeria Orso, Gabriella Francesca Amalia Pernice, Federico Corradini, Luca Fabbri, Luciano Gamberini, *Italy*

- **Research on Conveying User Experiences Through Digital Advertisement**

Stephanie Dwiputri Suciadi, Miwa Nakanishi, *Japan*

- **Development of a pen-type device for SPIDAR-tablet that presents force and thermal sensations**

Kaede Nohara, Seiya Sekiwa, Makoto Sato, Takehiko Yamaguchi, Tetsuya Harada, *Japan*

- **Development of SPIDAR-HMD for a standalone HMD**

Yoshiki Takahashi, Ryosuke Futatsumori, Makoto Sato, Takehiko Yamaguchi, Tetsuya Harada, *Japan*

- **An HMD-Integrated Haptic Device for Force, Friction, and Thermal Sensations of Fingertip**

Takahiro Ooshima, Ryuichi Osaki, Shimpei Matsukiyo, Makoto Sato, Takehiko Yamaguchi, Tetsuya Harada, *Japan*

- **Study on Virtual Reality Performance from the User's Individual Characteristics**

Daiji Kobayashi, Seiji Kikuchi, *Japan*

- **Motion Primitive Segmentation Based on Cognitive Model in VR-IADL**

Taisei Ando, Takehiko Yamaguchi, Norito Kohama, Maiko Sakamoto, *Japan*; Tania Giovannetti, *United States*; Tetsuya Harada, *Japan*

S061: Complex Information Environments

Chair(s): To be announced



- **Lessons Learned from Applying Requirements and Design Techniques in the Development of a Machine Learning System for Predicting Lawsuits Against Power Companies**
Luis Rivero, Carlos Portela, Jose Boaro, Pedro Santos, Venicius Rego, Geraldo Braz Junior, Anselmo Paiva, Erika Alves, Milton Oliveira, Renato Moraes, Marina Mendes, *Brazil*
- **Using User-Guided Development to Teach Complex Scientific Tasks Through a Graphical User Interface**
Alexis R. Tudor, Richard M. Plotkin, Aarran W. Shaw, Ava E. Covington, Sergiu Dascalu, *United States*
- **Impact of Task Cycle Pattern on Project Success in Software Crowdsourcing**
Razieh Saremi, *United States*; Marzieh Lotfalian Saremi, *Canada*; Sanam Jena, Robert Anzalone, Ahmed Bahabry, *United States*
- **Decision Support for Prolonged, and Tactical Combat Casualty Care**
Christopher Nemeth, Adam Amos-Binks, Natalie Keeney, Yuliya Pinevich, Gregory Rule, Dawn Laufersweiler, Isaac Flint, Vitaly Hereasevich, *United States*

Engineering Psychology and Cognitive Ergonomics

S062: Studies on Human Cognition and Mental Workload

Chair(s): Wenbin Guo, *United States*



Francis Quinby, Seyeon Kim, Marco Pollanen, Wesley Burr, Michael Reynolds, *Canada*

- **The Influence of Gender on Human's Cognitive Ability and the Correlation Research of Different Cognitive Dimensions**
Zhongqi Liu, Le Dong, Qianxiang Zhou, *P.R. China*
- **How Metacognitive Monitoring Feedback Influences Workload in a Location-based Augmented Reality Environment**
Wenbin Guo, Jung Hyup Kim, *United States*
- **Cognitive Activity Recognition based on Self-Supervised Learning from EEG Signals**
Yifeng Yang, Yingjie Zhao, Yanyu Lu, Shan Fu, *P.R. China*

Universal Access in Human-Computer Interaction

S063: New Technologies for Universal Access

Chair(s): Simeon Keates, *United Kingdom*

- **Contextual Cues: The Role of Machine Learning in Supporting Contextually Impaired Users**
Martin Kinch, Simeon Keates, *United Kingdom*
- **Lending an Artificial Eye: Beyond Evaluation of CV-based Assistive Systems for Visually Impaired People**
Fotis P. Kalaganis, Panagiotis Migkatzidis, Kostas Georgiadis, Elisavet Chatzilari, Spiros Nikolopoulos, Ioannis Kompatsiaris, *Greece*
- **Criteria for the Adoption of a Support Information System for People's Mobility**
Laura Burzagli, Pier Luigi Emiliani, *Italy*
- **A Study into Accessibility and Usability of Automated Teller Machines for Inclusiveness**
Patrizia Willi, Leandro Soares Guedes, Monica Landoni, *Switzerland*

S064: Recent Developments in Techniques and Technologies for Universal Accessibility

Chair(s): Frode Eika Sandnes, Terje Gjøsæter, *Norway*



- **Image Search Versus Text Search Revisited: A Simple Experiment using a Kahoot Quiz**
Signe Aanderaa Eide, Ana-Maria Poljac, Frode Eika Sandnes, *Norway*
- **Web Accessibility and Web Developer Attitudes towards Accessibility in Mozambique**
Suraj Gupta, Terje Gjørseter, *Norway*; G. Anthony Giannoumis, *United States*
- **Screen Reader Accessibility Study of Interactive Maps**
Sayed Kamrul Hasan, Terje Gjørseter, *Norway*
- **The Ecosystem's Involvement in the Appropriation Phase of Assistive Technology: Choice and Adjustment of Interaction Techniques**
Charline Calmels, Caroline Mercadier, Frédéric Vella, Antonio Serpa, Philippe Truillet, Nadine Vigouroux, *France*
- **Viva: A Virtual Assistant for the Visually Impaired**
Zeeshan Ahmed Pachodiwalé, Yugeshwari Brahmanekar, Neha Parakh, Dhruvil Patel, Magdalini Eirinaki, *United States*

S065: Human-Computer Interaction to Improve Quality of Life

Chair(s): Sophie Lepreux, Kathia Marçal De Oliveira, *France*





Yohan Guerrier, Janick Naveteur, Christophe Kolski, Francoise Anceaux, *France*

- **Multimodal Interaction Framework based on Firebase Real-Time Database**

Youssef Guedira, Jose Rouillard, *France*

- **Design of Digital Therapeutic Workshops for People with Alzheimer's Disease**

Anne-Marie Dery-Pinna, Alain Giboin, Philippe Renevier-Gonin, *France*

- **ParkinsonCom Project: Towards a Software Communication Tool for People with Parkinson's Disease**

Kathia Marçal De Oliveira, *France*; Elise Batsele, *Belgium*; Sophie Lepreux, *France*; Elise Buchet, *Belgium*; Christophe Kolski, *France*; Mathilde Boutiflat, *Belgium*; Veronique Delcroix, *France*; Hélène Geurts, *Belgium*; Kodzo Apedo, *France*; Loic Dehon, *Belgium*; Houcine Ezzedine, Yohan Guerrier, *France*; Marie-Claire Haelewyck, Nicolas Jura, *Belgium*; Philippe Pudlo, Yosra Rekik, *France*

- **Design and Validation of a Stress Detection Model for Use with a VR Based Interview Simulator for Autistic Young Adults**

Miroslava Migovich, Alex Korman, Joshua Wade, Nilanjan Sarkar, *United States*

- **Tasteful: A Cooking App Designed for Visually Impaired Users**

Yunran Ju, Zhenyu (Cheryl) Qian, Weilun Huang, *United States*

Cross-Cultural Design

S066: Design for Modern Society

Chair(s): Na Chen, *P.R. China*



- **Cross-cultural Design and Evaluation of Robot Prototypes based on Kawaii (Cute) Attributes**
Dave Berque, Hiroko Chiba, *United States*; Tipporn Laohakangvalvit, Michiko Ohkura, Peeraya Sripian, Midori Sugaya, *Japan*; Kevin Bautista, Jordyn Blakey, *United States*; Feng Chen, Wenkang Huang, Shun Imura, Kento Murayama, *Japan*; Eric Spehlmann, Cade Wright, *United States*
- **Towards Effective Robot-Assisted Photo Reminiscence: Personalizing Interactions through Visual Understanding and Inferring**
Edwinn Gamborino, Alberto Herrera Ruiz, Jing-Fen Wang, Tsung-Yuan Tseng, Su-Ling Yeh, Li-Chen Fu, *Taiwan**, *China*
- **Maintenance Feasibility Analysis Based on a Comprehensive Indicator**
Shaowen Ding, Teng Zhang, Changhua Jiang, Bo Wang, Yan Zhao, Fenggang Xu, Jianwei Niu, *P.R. China*
- **A Study for Evaluations of Automobile Digital Dashboard Layouts Based on Cognition Electroencephalogram**
Hao Yang, Jitao Zhang, Ruoyu Jia, *P.R. China*

S067: CCDHA -- Cross-Cultural Design for Humanity and Arts - II

Chair(s): Po-Hsien Lin, Rungtai Lin, *Taiwan**, *China*



- **Redesign, Transformation and Reflection of Local Marriage Custom Articles: A Case Study of "Ten-Mile Red Dowry"**
Jing Cao, *Taiwan**, *China*; Wai Kit NG, *Hong Kong*; Yuheng Tao, *P.R. China*; Po-Hsien Lin, *Taiwan**, *China*
- **A Study on the Cognitive Differences of Incomplete Beauty in Sculptures among Audiences**
Yuheng Tao, HongQian Zheng, *P.R. China*; Jing Cao, Po-Hsien Lin, *Taiwan**, *China*
- **On the Zen Connotation in Product Design**
Tze-Fei Huang, Po-Hsien Lin, *Taiwan**, *China*
- **Discussing how the Viewer, the Author, and the Work See and be Seen in the Photography Works of "Farewell to the Island"**
HongQian Zheng, Yuheng Tao, *P.R. China*; Rungtai Lin, Po-Hsien Lin, *Taiwan**, *China*
- **Inheritance Model and Innovative Design of Chinese Southern Ivory Carving Culture and Craft**
Ya Juan Gao, Hao Wu Chen, Min Ling Huang, *P.R. China*; Rungtai Lin, *Taiwan**, *China*
- **From Imitation to Innovation: A Case Study of An Amateur Artist**
Rungtai Lin, *Taiwan**, *China*; Yikang Sun, *P.R. China*; Andrew Yu, *United States*

Social Computing and Social Media

S068: The World of Consumers and Social Media: People and Organizations

Chair(s): Adela Coman, *Romania*



- **The Effect of Social Media based Electronic Word of Mouth on Propensity to buy Wearable Devices**
David Ntumba, Adheesh Budree, *South Africa*
- **Social Media, Leadership and Organisational Culture: The Case of Romanian Leaders**
Adela Coman, Ana-Maria Grigore, Andreea Ardelean, *Romania*
- **First Steps in The Entrepreneurial Practice of Marketing**
Cristian G. Popescu, Raluca Ion, *Romania*
- **The impact of Social Media Marketing on Impulse Buying**
Adheesh Budree, Warren Driver, Abongile Pandle, Gandidzanwa Tanaka, *South Africa*

S069: Multimodal Social Signal Processing and Applications

Chair(s): Hung-Hsuan Huang, Shogo Okada, Ryo Ishii, *Japan*



- **Analysis of Modality-based Presentation Skills Using Sequential Models**
Su Shwe Yi Tun, Shogo Okada, Hung-Hsuan Huang, *Japan*; Chee Wee Leong, *United States*
- **Estimating Interaction State from Nonverbal Cues and Utterance Events: A Preliminary Study to Support Ideation Facilitation in Living Lab**
Ichiro Umata, Sumaru Niida, Koki Ijuin, Tsuneo Kato, Seiichi Yamamoto, *Japan*
- **Multimodal Emotion Analysis based on Acoustic and Linguistic Features of the Voice**
Leon Koren, Tomislav Stipancic, *Croatia*
- **Estimation of Empathy Skill Level and Personal Traits using Gaze Behavior and Dialogue Act during Turn-changing**
Ryo Ishii, Shiro Kumano, Ryuichiro Higashinaka, Shiro Ozawa, Tetsuya Kinebuchi, *Japan*
- **Activity Comparison of the Participants using Japanese as L2 and their L1 in Group Discussion**
Hung-Hsuan Huang, Zi-Yu Pei, Kazuhiro Kuwabara, *Japan*

Digital Human Modeling and Applications in Health, Safety, Ergonomics and Risk Management

S070: AHP and Applications in Decision Making

Chair(s): Genett Jimenez-Delgado, *Colombia*



José Carlos De Sá, Joni Pinto Jorge, Gilberto Santos, Maria João Félix, Luis Barreto, *Portugal*; Genett Jimenez-Delgado, Carlos Rondón, Carlos Vargas-Mercado, *Colombia*

- **An Integrated Framework based on Fuzzy AHP-TOPSIS and Multiple Correspondences Analysis (MCA) for Evaluate the Technological Conditions of the Tele-worker in Times of Pandemic: A Case Study**

Leonel Hernandez-Collantes, Nidia Balmaceda-Castro, Jiseth Guerra-Renard, Ana Charris-Muñoz, Lorayne Solano-Naizzir, Carlos Vargas-Mercado, Daniel Alcazar-Franco, *Colombia*

- **A FAHP-VIKOR Approach for Supporting the Selection of Tomography Equipment in LMIC hospitals: A Case Study**

Miguel Angel Ortiz Barrios, Natalia Jaramillo Rueda, *Colombia*; Antonella Petrillo, *Italy*; Zaury Fernández-Mendoza, Lucelys Vidal-Pacheco, *Colombia*

- **Implementation of Lean Six Sigma to Lessen Waiting Times in Public Emergency Care Networks: A Case Study**

Miguel Angel Ortiz Barrios, Dayana Coba-Blanco, Genett Jimenez-Delgado, *Colombia*; Valerio Salomon, *Brazil*; Pedro Vicente Lopez-Meza, *Colombia*

S071: Ergonomics and Safety in Transport and Industry

Chair(s): Vincent G. Duffy, *United States*



- **A Systematic Literature Review on the Interaction between COVID-19 and Transportation**
Yan Chen, Vincent G. Duffy, *United States*
- **Safety Management and Challenges Associated with Industry 4.0 on Transportation and Logistics: A Systematic Literature Review**
Chien-Hsien Lin, *Taiwan**, *China*; Vincent G. Duffy, *United States*
- **Bibliometric Analysis on the Safety of Autonomous Vehicles with Artificial Intelligence**
Hak Jun Kim, Vincent G. Duffy, *United States*
- **Injury Prevention for Transportation Safety: A Bibliometric Analysis**
Ali Alkhaleefah, Quirinus Renardo, Vincent G. Duffy, *United States*
- **Trust and Automation: A Systematic Review and Bibliometric Analysis**
Zhengming Zhang, *P.R. China*; Vincent G. Duffy, Renran Tian, *United States*

Design, User Experience, and Usability

S072: Fashion and the Digital Transformation: HCI Perspectives

Chair(s): Lorenzo Cantoni, Nadzeya Kalbaska, *Switzerland*





- **Online Communication Design within Fashion Curricula**
Anna Picco-Schwendener, Tekila Harley Nobile, Tetteng Gaduel Thaloka, *Switzerland*
- **Eco-Activism, Human-Computer Interaction and Fast Fashion**
Antonio Nucci, Matthew Hibberd, *Switzerland*
- **Exploration of Norms and Policies in Digital Fashion Domain Using Semantic Web Technologies**
Soheil Roshankish, Nicoletta Fornara, *Switzerland*
- **COVID-19 Outbreak and Fashion Communication Strategies on Instagram: A Content Analysis**
Alice Noris, Lorenzo Cantoni, *Switzerland*
- **Participatory Design to Create Digital Technologies for Batik Intangible Cultural Heritage. The case of iWareBatik**
Puspita Ayu Permatasari, *Indonesia*; Lorenzo Cantoni, *Switzerland*

S073: UX Aspects in Product Design

Chair(s): Nan Zhang, *P.R. China*



- **Modular Approach to Designing 3D Printed Products: Custom HCI Design and Fabrication of Functional Products**
Robert Phillips, James Tooze, Paul Smith, Sharon Baurley, *United Kingdom*
- **A Case Study of Augmented Physical Interface by Foot Access with 3D Printed Attachment**
Tatsuya Minagawa, Yoichi Ochiai, *Japan*
- **Research on Intelligent Classified Trash Can and Smart Application Design—Achieving Green Smart Home Living in China**
Nan Zhang, Yanlin Wu, Kong Yingao, Jinsong Lv, *P.R. China*
- **Lacquer Jewelry Design of Shanghai Style Based on User Perception Preference**
Yalan Yu, RongRong Fu, *P.R. China*
- **Product Interventions and User Performance: Implications for Public Design to Achieve Sustainable Practice**
Ming Jun Luo, Jia Xin Xiao, Wenhua Li, *P.R. China*

S074: Developments in DUXU: Contributions for the Context of Technological Environments - II

Chair(s): Freddy Paz, *Peru*



- **A Framework Based on UCD and Scrum for the Software Development Process**
Daniela Argumanis, Arturo Moquillaza, Freddy Paz, *Peru*
- **Developing and Validating a Set of Usability and Security Metrics for ATM Interfaces**
Fiorella Falconi, *United States*; Arturo Moquillaza, Joel Aguirre, Freddy Paz, *Peru*
- **Heuristic Evaluation of Android-based Applications with Multiple Screen Sizes Support: A Case Study**
Juan Carlos Romaina, *Peru*
- **A Usability Evaluation Process Proposal for ATM Interfaces**
Joe Sahua, Arturo Moquillaza, Freddy Paz, *Peru*
- **Improving the Withdrawal Functionality on ATM Using a UCD Framework - A Case Study**
Joel Aguirre, *Peru*; Fiorella Falconi, *United States*; Rodrigo Serrano, Arturo Moquillaza, Freddy Paz, *Peru*

Learning and Collaboration Technologies

S075: Interactive Learning Ecosystems - III

Chair(s): David Fonseca, Francisco J. García-Peñalvo, *Spain*





Ayako Suto, Koichi Nishiyama, *Japan*

- **Designing Learning Environments in a Digital Time – Experiences in two Different Subjects at NTNU, Norway Autumn 2020**

Tord Mjøsund Talmo, Øystein Marøy, Helene Røli Karlsen, *Norway*

- **Evaluation Design for Learning with Mixed Reality in Mining Education based on a Literature Review**

Stefan Thurner, Lea M. Daling, *Germany*; Markus Ebner, Martin Ebner, Sandra Schön, *Austria*

- **Nudge for Note Taking Assist System: A Learning Strategy Feedback System Among Learners Through Their Tablet**

Takaki Kondo, Kyoichi Yokoyama, Tadashi Misono, Rieko Inaba, Yuki Watanabe, *Japan*

- **Create Children's Programming Teaching Aids with Chinese Characteristics: Provide A Method Framework to Assist Designers in Designing**

Zhijuan Zhu, Xinjun Miao, Yan Qin, Wenzhen Pan, *P.R. China*

Human Aspects of IT for the Aged Population

S076: Designing for the Elderly - II

Chair(s): To be announced



- **Towards Representation of Daily Living Activities by Reusing ICF Categories**
Satoshi Nishimura, Ken Fukuda, *Japan*
- **Application of Sustainability Design Method for Well-Being Design**
Yi Li, *P.R. China*
- **Research on the Design of Smart Bracelets for the Elderly based on Maslow's Hierarchy of Needs**
TianYu Shi, Yu Wei, *P.R. China*
- **Interface Adaption to Elderly Users: Effects of Icon Styles and Semantic Distance**
Kairu Zhao, Xiaochun Wang, Linyao Bai, *P.R. China*

HCI in Games

S077: Recent Developments in Gaming and HCI - II

Chair(s): Khaldoun Dhou, *United States*; Barbara Caci, *Italy*

- **An Exploration of the Fear of Attack Strategy in Chess and its Influence on Class-A Players of Different Chess Personalities: An Exploration Using Virtual Humans**
Khaldoun Dhou, *United States*
- **Game Design, Creativity and e-Learning: The Challenges of Beginner Level Immersive Language Learning Games**
Tiago Barros Pontes e Silva, Raquel Pereira Pacheco, Mariana Da Silva Lima, Maurício Miranda Sarmet, Maria Luísa De Carvalho Cascelli de Azevedo, Vitor Henrique Malcher Ferreira, Carla Denise Castanho, *Brazil*
- **Virtual Reality for Rich Interaction with Cultural Heritage Sites**
Nouf AlMuraikhi, Fatima Almalki, Fadeela AlDahnim, Osama Halabi, *Qatar*
- **Exploring Coordination Patterns in VR-based Rehabilitation for Stroke using the Kinect Sensor**
Maria F. Montoya, Julian F. Villada, *Colombia*; John Muñoz, *Canada*; Oscar A. Henao, *Colombia*





Chair(s): Lena Levin, *Sweden*

- **Gender, Smart Mobility and COVID-19**
Angela Carboni, *Italy*; Mariana Costa, Sophia Kalakou, *Portugal*; Miriam Pirra, *Italy*
 - **Smart and Inclusive Bicycling? Non-users' Experience of Bike-Sharing Schemes in Scandinavia**
Michala Hvidt Breengaard, *Denmark*; Malin Henriksson, Anna Wallsten, *Sweden*
 - **Plotting the nexus between Inclusive Mobilities, Climate Mitigation Policies, Covid-19 outbreak and Digitalisation. Highlights from Norway.**
Tanu Priya Uteng, Lars Bocker, *Norway*
 - **Smart is (not) always Digital! Expanding the Concept of Assistive Technology: The Roller as an Age-Based, Gendered and Social Innovation**
Hilda Roemer Christensen, *Denmark*
 - **Smart Mobility for all – TInnGO provides a new tool for gender and diversity action planning (GaDAP)**
Lena Levin, *Sweden*
-

Adaptive Instructional Systems

S079: Essentials of Learner Modeling for Adaptive Instructional Systems (AISs)

Chair(s): Ani Grubisic, *Croatia*



- **Enhancing Personalization by Integrating Top-down and Bottom-up Approaches to Learner Modeling**
Diego Zapata-Rivera, Burcu Arslan, *United States*
- **Modelling and Quantifying Learner Motivation for Adaptive Systems: Current Insight and Future Perspectives**
Fidelia A. Orji, Julita Vassileva, *Canada*
- **Croatian POS Tagger as a Prerequisite for Knowledge Extraction in Intelligent Tutoring Systems**
Daniel Vasić, *Bosnia and Herzegovina*; Branko Žitko, Ani Grubisic, Slavomir Stankov, Angelina Gašpar, Ines Šarić-Grgić, Suzana Tomaš, Ivan Peraić, *Croatia*; Matea Markić-Vučić, *Bosnia and Herzegovina*

Design, Operation and Evaluation of Mobile Communications

S080: Mobile Communications Applications from Users' Perspectives

Chair(s): June Wei, *United States*

- **The Influence Mechanism of Demand Fluctuation on Service Quality of Digital Supply Chain: the Moderating Effect of Digital Technology Capability**
Caihong Liu, *P.R. China*; Hannah Ji, June Wei, *United States*
- **Bibliometric Structured Review of Mobile Information Systems**
Sunday Adewale Olaleye, Ismaila Temitayo Sanusi, Oluwaseun Alexander Dada, *Finland*
- **The Use of mPOS in Mexico**
Sunday Adewale Olaleye, *Finland*; Manuela Gutiérrez-Leefmans, *Mexico*

Culture and Computing

S081: Digital Memory Modalities - Inquiring the Role of HCI for Participatory Memory Practices

Chair(s): Gertraud Koch, *Germany*





- **Infrastructuring for Collective Heritage Knowledge Production**
Sanna Marttila, *Denmark*; Andrea Botero, *Finland*
- **Breaking Boundaries, Creating Connectivities: Enabling Access to Digitized Museum Collections**
Cassandra Kist, *United Kingdom*; Quoc-Tan Tran, *Germany*
- **Youth and Algorithmic Memory: Co-Producing Personal Memory on Instagram**
Jennifer Krueckeberg, *Germany*
- **On Complexity of GLAMs' Digital Ecosystem: APIs as Change Makers for Opening Up Knowledge**
Angeliki Tzouganatou, *Germany*
- **Civic Community Archiving with the Platform for Experimental Collaborative Ethnography: Double Binds and Design Challenges**
Kim Fortun, Mike Fortun, Angela Hitomi Skye Crandall Okune, *United States*; Tim Schütz, *Germany*; Shan-Ya Su, *Taiwan**, *China*

S082: Digital Technologies and Heritage Conservation - II

Chair(s): Claudia Trillo, Busisiwe Chikomborero Ncube Makore, *United Kingdom*





Rania Aburamadan, *Jordan*; Athena Moustaka, Claudia Trillo, Busisiwe Chikomborero Ncube Makore, Chika Udeaja, Kwasi Gyau Baffour Awuah, *United Kingdom*

- **The MaDiH (مديح): Mapping Digital Cultural Heritage in Jordan, Opportunities and Limitations.**
Fadi Bala'awi, Shatha Mubaideen, *Jordan*; James Smithies, Pascal Flohr, Alessandra Esposito, *United Kingdom*; Carol Palmer, Sahar Idwan, *Jordan*
- **The Circular City Implementation: Cultural Heritage and Digital Technology**
Martina Bosone, Francesca Nocca, Luigi Fusco Girard, *Italy*
- **Social Media Data for the Conservation of Historic Urban Landscapes: Prospects and Challenges**
Manal Ginzarly, *Lebanon*

Artificial Intelligence in HCI

S083: AI and HCI for Application Domains of Societal Impact - II

Chair(s): Sandeep Kaur Kuttal, *United States*

- **Why Developing Simulation Capabilities Promotes Sustainable Adaptation to Climate Change**
Gabriele De Luca, Thomas J. Lampoltshammer, *Austria*; Shahanaz Parven, *Russia*
- **Toward AI-Based Scenario Management for Cyber Range Training**
Jo Hannay, Audun Stolpe, Muhammad Mudassar Yamin, *Norway*
- **DeepDive: The Use of Virtual Worlds to Create an Ethnography of an Ancient Civilization**
Thomas Palazzolo, Ashley Lemke, Chencheng Zhang, Sarah Saad, Robert Reynolds, John O'Shea, *United States*
- **Genetic Algorithm to Plan Routes - Case: Waste Collectors - Huanuco, Peru 2020 Genetic Algorithm To Plan Routes. Case: Waste Collectors - Huanuco, Peru. 2020**
Julissa Elizabeth Reyna-González, *Peru*; Alberto Gomez Fuertes, *Spain*; Moisés David Reyes Pérez, Gilberto Carrión-Barco, César Augusto Piscocoy Vargas, *Peru*





* Consistent with the 'one-China policy' of the United Nations Organization and the United States (maintaining unofficial relations with Taiwan after 1979).

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[Conference Administration administration@hcii2021.org \(mailto:administration@hcii2021.org\)](mailto:administration@hcii2021.org) |
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